**Medicine, Dentistry and Medical laboratory**

# Cell Biology

**Lecture 3**

## Cell structure & function - 1

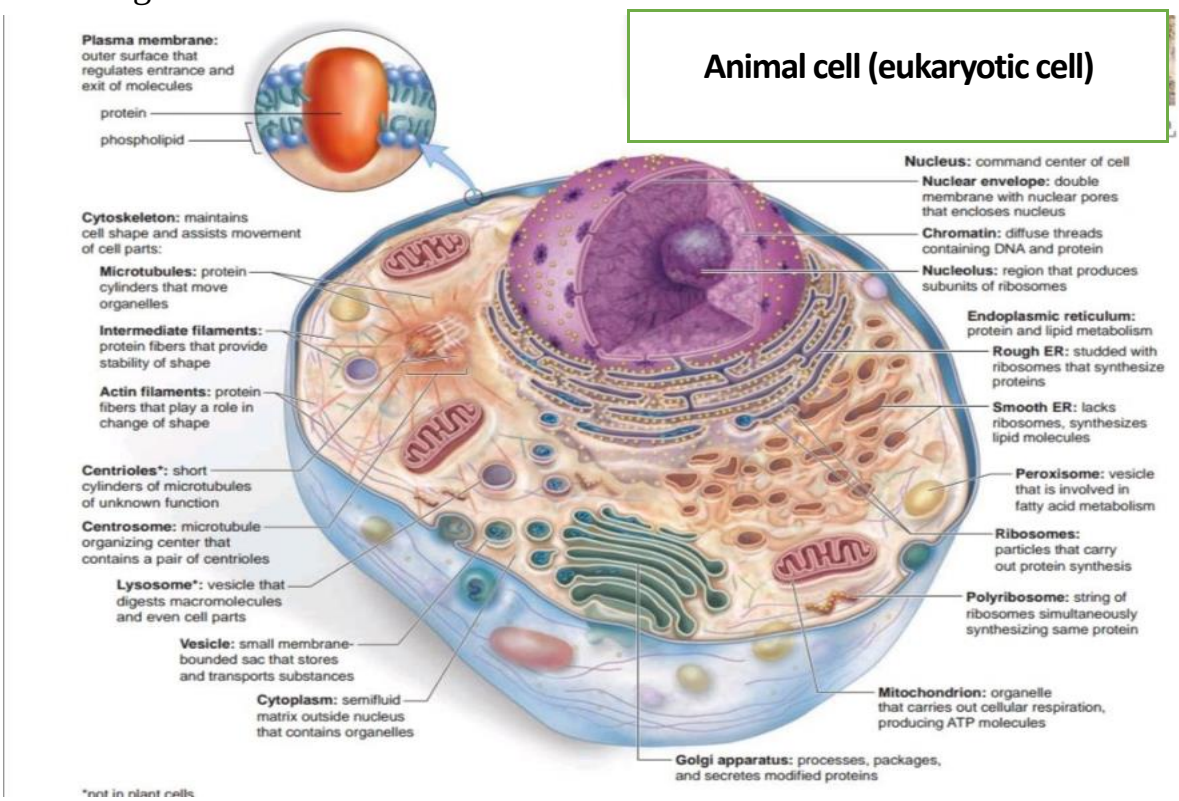
## A general living cell (e.g. animal cell) is composed of:

1- Cell surface membrane (plasma membrane).

2- Cytoplasm.

3- Cytoskeleton.

4- Cell organelles.




## 1- Cell membrane:

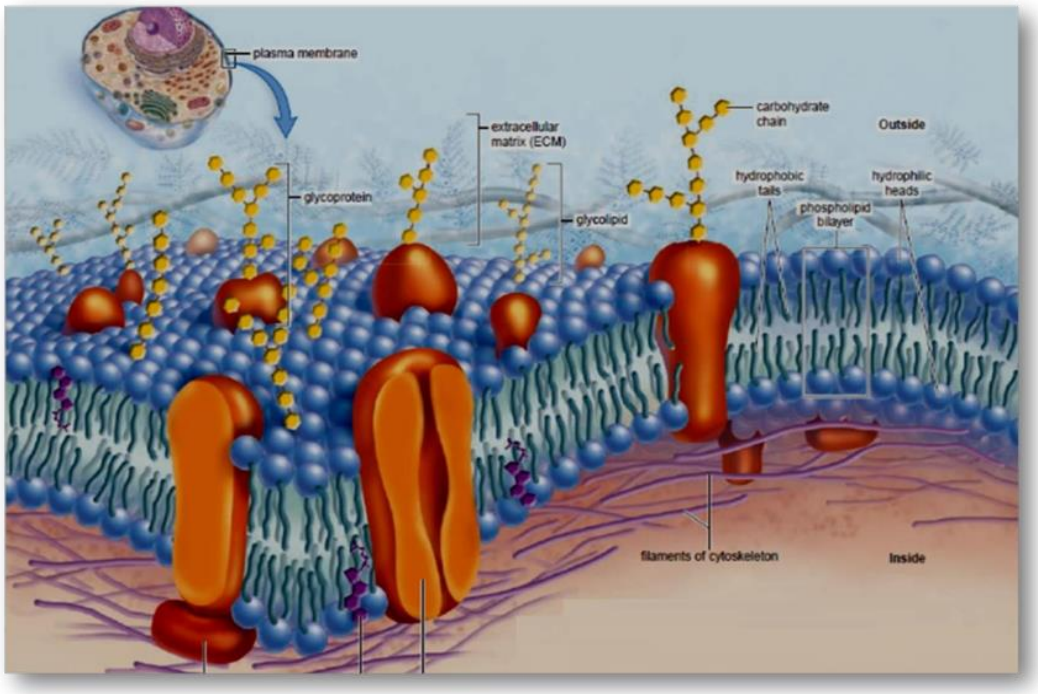
It consists of:

1- Phospholipid bilayer

2- Proteins

3- Carbohydrate chains

4- Lipid (cholesterol molecules).




**Cell membrane.**

*** The plasma membrane** keeps the organelles and the cytoplasm inside the cell and regulates the passage of molecules into and out of the cell (obtain nutrients, gain information and rid waste).

***** You will have more details about cell membrane structure and function in lecture 5.

## 2- Cytoplasm (cytosol):

Is a semi fluid medium which holds structures known as cell organelles.

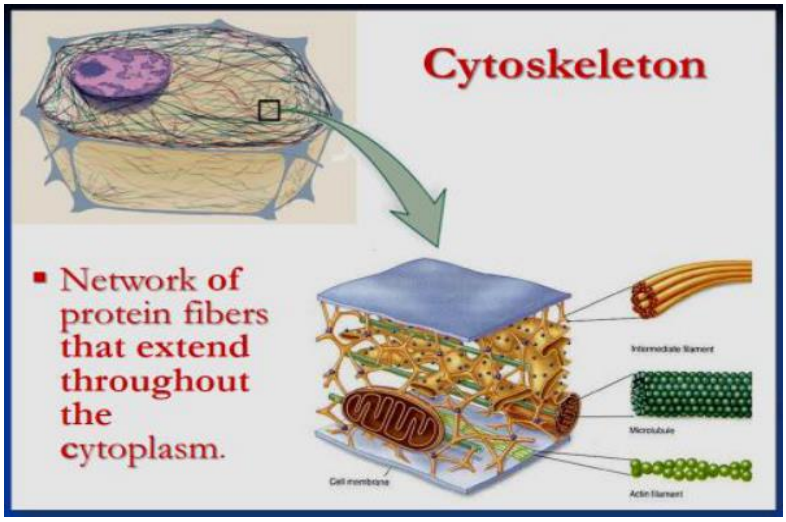
* Cytoplasm + Nucleus = Protoplasm (living material).

* The cytoplasm contains about 90% water, small molecules (e.g. ions, sugars, vitamins etc...) large molecules (e.g. proteins, fats) which form a colloidal solutions.

* The colloidal solution is the solution in which the solute molecules are relatively large.

**3- The cytoskeleton:**

The protein components of the cytoskeleton interconnect and extend from the nucleus to the plasma membrane in eukaryotic cells. It contains actin filaments, intermediate filaments, and microtubules.




**Function:**

* The cytoskeleton maintains cell shape and allow the cell and its organelles to move.

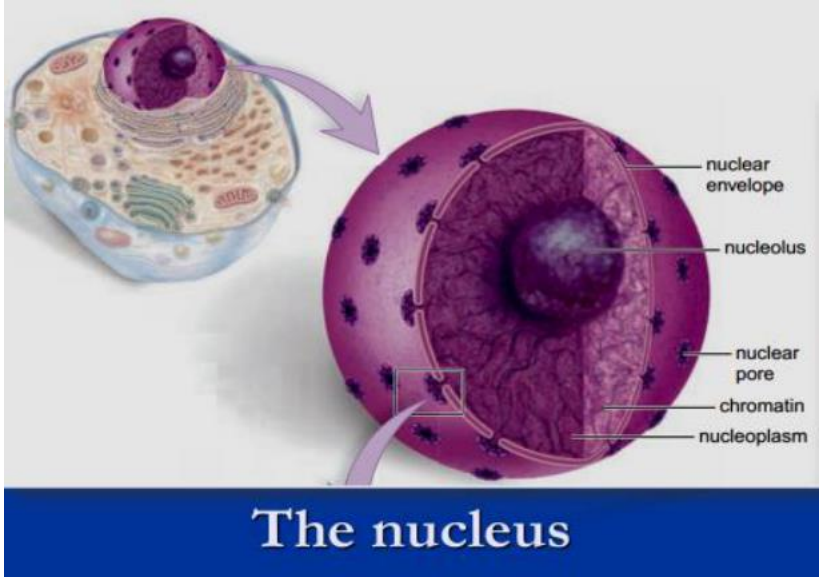
* The cytoskeleton is often compared to the bones and muscles of an animal.

**4- Cell organelles:**

Animal cell include: nucleus, endoplasmic reticulum, ribosomes, Golgi apparatus, lysosomes, mitochondria, centrioles, cilia and flagella, vesicles, vacuoles and peroxisomes.

**1- The nucleus:**

The largest cellular organelle. It may not found in some types of cells (e.g. mature red blood cells).




**It located near the center of most cells and consists of:**

1- Nuclear envelope. 3- Chromatin.

2- Nuclear pores. 4- Nucleolus.

## 1- Nuclear envelope:

***** The nucleus is separated from the cytoplasm by a double membrane known as the nuclear envelope. Even so, the nucleus communicates with the cytoplasm.

## 2- Nuclear pores:

The nuclear envelope contains pores lined by a complex of eight proteins.

***** Nuclear pores serve as passageways for substances to pass into and out of the nucleus.

## 3- Chromatin:

It looks grainy, but actually it is a network of strands that condenses and undergoes coiling into rod-like structures called chromosomes just before the cell divides.

***** The chromosomes are the carriers of genetic information.
It contains DNA, protein, and some RNA (ribonucleic acid). Genes, composed of DNA, are units of heredity located on the chromosomes.

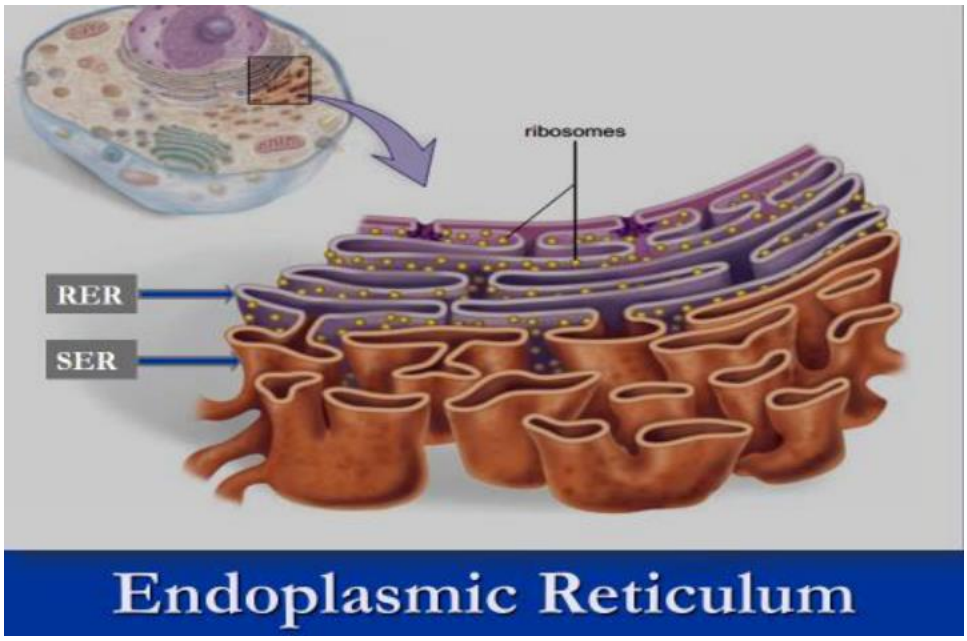
**4-The nucleolus** is a region of chromatin where ribosomal RNA is produced, and ribosomal subunits are assembled.

**Function**:

* The nucleus contains the genetic information that is passed on from cell to cell and from generation to generation.
* Control all cell activities.

## 2- Endoplasmic reticulum:

It consisting of a complicated system of membranous channels and saccules (flattened vesicles), is physically continuous with the nuclear envelope.




* The Endoplasmic reticulum (**ER**) consists of:
1- Rough ER (**RER**).
2- Smooth ER (**SER**).

1- **RER** is studded with ribosomes on the side of the membrane that faces the cytoplasm.

***** The number of ribosomes in a cell varies depending on its functions.

***** Ribosomes are particles where protein synthesis occurs. They are composed of small and large subunits. Each subunit has its own mix of proteins and RNA (ribonucleic acid).

**Function:**

***** It has the capacity to produce proteins and glycoproteins (protein synthesis).

**2- SER**, which is continuous with rough ER, does not have attached ribosomes.

***** Its function depends on the organ.

**Function:**

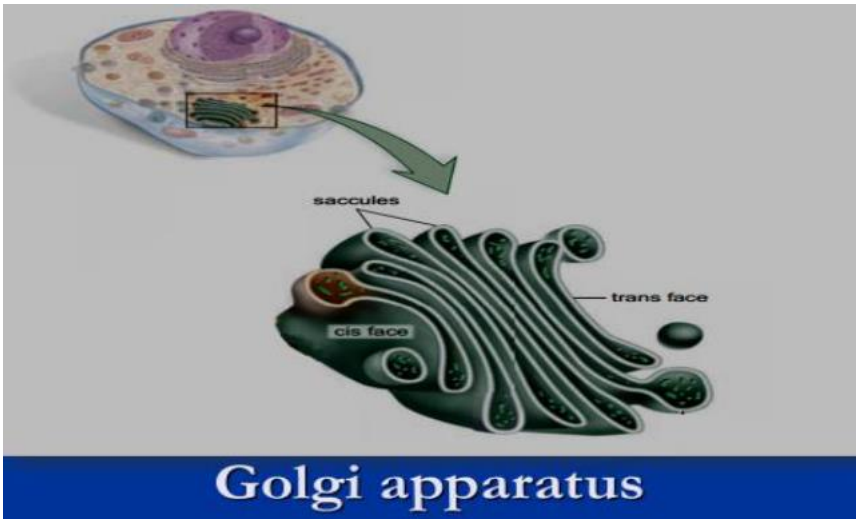
***** In some organs, SER is associated with the production of lipids.

***** In the testes, SER produces testosterone, a steroid hormone.

***** In the liver, SER helps detoxify drugs.

**3- Golgi apparatus**:

It is a group of flattened membranous sacs which form vesicles at one end and are completely formed at the other (stack of flattened curved saccules).



**Function:**

***** It processes proteins and lipids and packages them in transport vesicles that distribute these molecules to various locations.

*It receives transport vesicles from the SER, then it modifies, sorts and packages proteins for use in cell or to be secreted from the cell.

* Sometimes involved in secretion of carbohydrates as in the synthesis of new cell walls in plants.
* Formation of lysosomes.